Pengxu (Vincent) Chen

Structured Problem-Solving | Quantitative Analysis | Communication
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Meet me in 60 seconds

PROFILE

Award-winning PhD researcher who turns complex, data-heavy questions into clear analyses and practical recommendations. Has led projects involving national supercomputer, managed research budget of £160,000, and explained technical findings to non-specialists. Seeking to use analytical skills to solve practical problems.

EDUCATION

The University of Edinburgh — Edinburgh, UK
PhD in Engineering (Applied Physics, fully funded studentship) — *2022 – 2026*

Zhejiang University (3rd in China, Times Higher Education 2027) — Hangzhou, China
MEng in Engineering Materials; Grade: 85.7/100 — *2019 – 2022*

China University of Geosciences — Wuhan, China
BEng in Structural Engineering; Grade: 88.4/100; Rank: 2/33 — *2015 – 2019*

RESEARCH & PROJECT EXPERIENCE

The University of Edinburgh — Edinburgh, UK
PhD Researcher – Computational Modelling & Analysis — *09/2022 – 09/2026*

- Ran 3 concurrent doctoral projects and designed controlled comparisons across three water systems to test competing explanations for contrasting freezing responses; synthesised the results into one coherent research programme that provides guidance for industrial material and device design.
- Cut job-submission time by about 80% through automation and used 800,000 compute units (worth £160,000) on the UK national supercomputer.
- Packaged my PhD quantitative analysis into documented codes and protocols on GitHub so other researchers can reproduce the results and reuse the workflow without my involvement.
- Delivered 6 international conference talks, published 2 first-author papers, and contributed to a national supercomputing case study on the societal value of freeze desalination, for audiences outside the field.

University of Basel — Basel, Switzerland (Remote)
Research Collaborator – Computational Research — *04/2026 – 08/2026*

- Delivered a repeatable modelling workflow in an unfamiliar field (biological materials) by quickly adopting a new software platform within an international collaboration.

Zhejiang University — Hangzhou, China
MEng Researcher – Materials Modelling & Research Infrastructure — *09/2019 – 06/2022*

- Led a subtask in a national highway-safety R&D programme: evaluated computing options, built the budget and schedule case, and won approval from the national computing provider (January 2022).
- Designed the structure of a shared pavement research database around researchers' requirements and coordinated implementation with an external provider. Collected laboratory, field and other research data for the database, which was used by approximately 15 researchers.
- Explained pavement-material strength by combining modelling, engineering theory and experiments; validated the model and published 2 first-author papers.

LEADERSHIP & COMMUNICATION

- **Team leadership:** Led a five-member team in an international transport-infrastructure simulation competition (entry, research plan, final report), earning an Honourable Mention; previously led a three-student anti-icing pavement coating project from proposal to final report.
- **Stakeholder coordination:** Coordinated participant communications and guest logistics for TRC 2021 (approximately 200 attendees), independently managing invitation emails.
- **Mentoring & teaching:** Mentored undergraduate and master's thesis students (the undergraduate mentee graduated with a First-Class degree); taught tutorials and lab demonstrations for three years.
- **Visual communication:** Turned complex research into visuals for multidisciplinary and public audiences, winning Best Poster (1 of 92) and a UK-wide Best Early Career award (1 of 30).

RECOGNITION & OUTPUTS

- 8 peer-reviewed articles (4 first-author) in top journals across disciplines; received 398 citations (October 2026).
- Best Early Career Researcher, UK national HPC Image and Video Competition (2025); Best Poster, University of Edinburgh School of Engineering (2024); UK national HPC case study (2026).
- President Scholarship (top 3%, 2018) and Talent Scholarship (top 0.2%, 2016), China University of Geosciences.

SKILLS & LANGUAGES

- **Tools:** Excel (advanced), PowerPoint (advanced), C++, MATLAB, Linux, HPC computing.
- **Languages:** English (fluent), Chinese (native), French (intermediate).